Assignment 2: Impact of Transformer Component Choices

Introduction

Attention is the core component of transformers. “Attention Is All You Need” (Vaswani et al., 2017) replaced recurrent processing with attention, sparking a revolution in deep learning leading to the large-language models of today. Hence understanding attention is key to understanding transformers and large-language models and remains an active area of research and development. Developments to attention include sparse attention, multi-query/grouped attention, flash attention and paged attention (Zhao et al., 2023), often aiming to address the problem of the quadratic complexity of computation.

Other areas of research since the introduction of the Transformer have investigated the representational characteristics of the attention mechanism, using techniques such as head diversity (Audhkhasi et al., 2022), ablation (Voita et al.,2019), head characterisation (Vig, 2019) and spectral analysis (Amsell et al. 2024). Multi-headed attention is thought to allow for different representational subspaces to be attended to across a sequence, whereas single-headed attention allows for only one subspace (Vaswani et al., 2017). Transformers are often viewed as black boxes – analysis of structural and representational qualities sheds light on internal behaviour, contributing to interpretability. Analysing head redundancy and dimensionality can also contribute to improvements in computational efficiency.

This assignment explores how varying the number of heads and layer depth in the attention mechanism affects model behaviour, using these research techniques to examine the structural properties of attention.

Methodology

I implement a custom transformer based on the architecture presented by Vaswani et al. (2017) using Rush et al.(2022)’s source code as reference. An encoder-only transformer is used with a classification head, as opposed to the larger encoder-decoder model they detail. A model width (d_model) of 32 and a feed-forward network size of 64 is defined. Building a “toy” transformer from scratch enables direct access to all elements allowing for easier structural analysis and reduces training time for fast experiments.

Dataset choice was also influenced by this approach. I experimented with several datasets including the Text REtrieval Conference (TREC) Question Classification dataset (Hugging Face, 2025), as well as the Tiny Shakespeare dataset (Karpathy, 2015). In the final implementation I use a subset (10000 reviews) of the IMDB Dataset of 50k Movie Reviews (Maas et al., 2011) to perform a classification task. For each review there is a binary sentiment, positive (1) or negative (0). The reviews range from 4 to 2450 words, averaging 227 words (Malik, 2025). The IMDB dataset is chosen as the length of the reviews provides longer range dependencies for the input sequences, whereas the TREC dataset generally features shorter sequence lengths, which are less likely to benefit from multiple heads or layers.

For the experiments, I test the performance of four attention mechanism variants - a single-head, single-layer model, a multi-headed model with 4 heads (single layer), a multi-headed model with 8 heads (single layer) and a multi-headed model with 4 heads and 2 layers. I train each variant on three seeds. Setting the seem ensures the same random values are sampled

from the Xavier uniform distribution for weight initialisation prior to training (Geeks for Geeks, 2025).

Results & Analysis

Loss (cross-entropy) and accuracy metrics for are stored for all variants for both training and validation data (Figures 1 and 2, Table 1) over the 8 epochs trained. Single-head and 4-head model variations perform the best in terms of accuracy, with the 4-head version performing best on validation accuracy after 3 epochs. Above 4 heads, accuracy drops as model size increases.

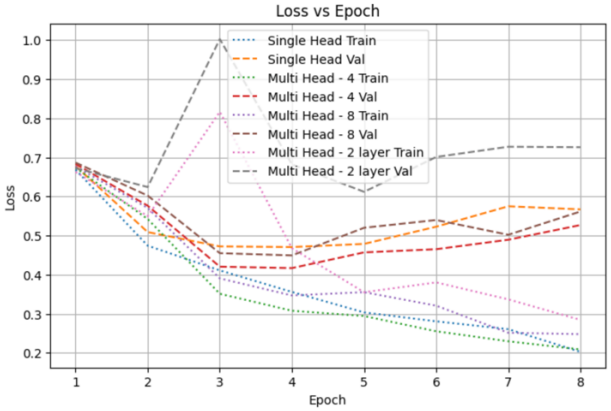


Figure 1 - Training accuracy for all model variations.

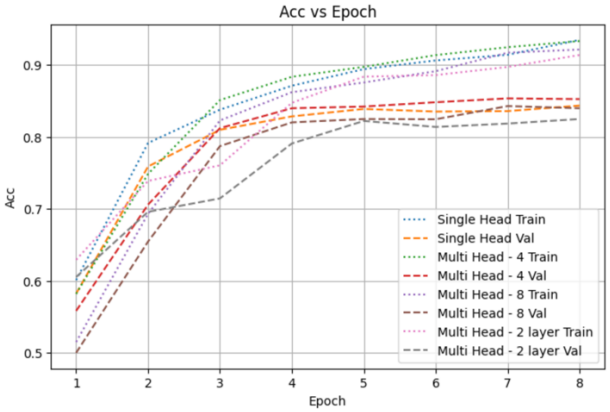


Figure 2 – Training Loss for all model variations.

Model variation (mean across all seeds)	**Training Accuracy** (max across epochs)	**Validation Accuracy** (max across epochs)
Single head	0.935	0.851
Multi-head (4), 1 layer	0.934	0.854
Multi-head (8), 1 layer	0.922	0.846
Multi- head (4), 2 layer	0.914	0.828

Table 1 – Accuracy results for all model variations

I utilise the custom model architecture to gather data on the representational properties of the attention head. I examine **head characteristics**, **cosine similarity** between head attention matrices, **singular values** and **rank** to observe behavioural differences across model variants.

Pande et al. (2021) identifies different head functions - **positional** (attends to adjacent tokens), **syntactic** (attends to tokens with a syntactic relation), **interrogative** (attends to a question mark) and **rare words** (attends to least frequent tokens). Plots of attention matrices from each head from each model for various sub-samples of text from the validation data, provide examples of these different head functions (Figures 3, 4 and 5) across variants.

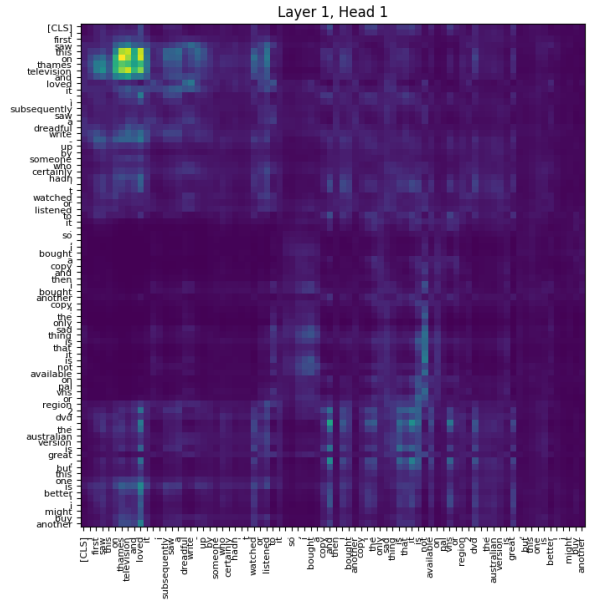


Figure 3 – Attention head with positional function

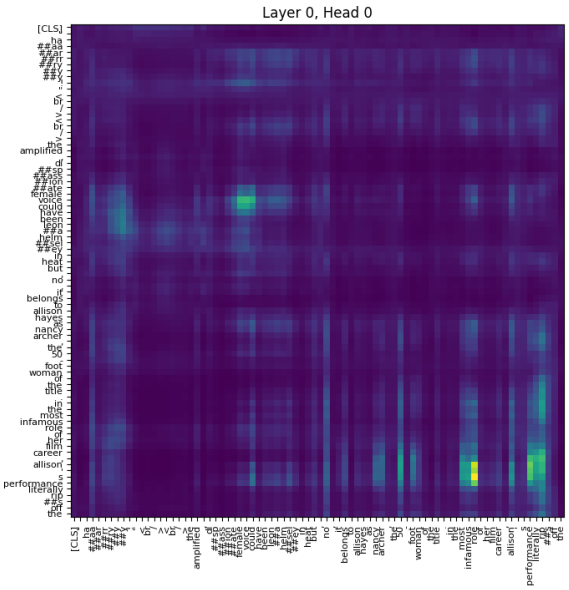


Figure 4 - Attention head with syntactic function

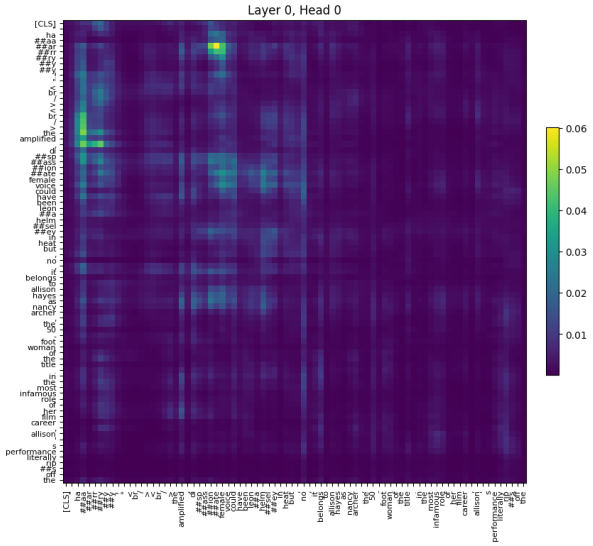


Figure 5 – Attention head with rare word function

Analysing head functions and visualising attention matrices offers insight into the comparable performance of single and multi-head attention variants. In Example 1- a negative review, the single-headed model attention head (Figure 6) finds a similar pattern to Head 0 in the multi-head (4) model (Figure 7), focusing on the relationship of the word “waste” to other related tokens in the text. For binary sentiment tasks, certain words may be highly informative features, allowing the model to predict sentiment without longer range dependencies, creating redundancy in multi-head architectures. Supporting this, the similarity between heads 0 and 3 (Figure 7,10) suggests these heads may be partially redundant, capturing overlapping features in the input.

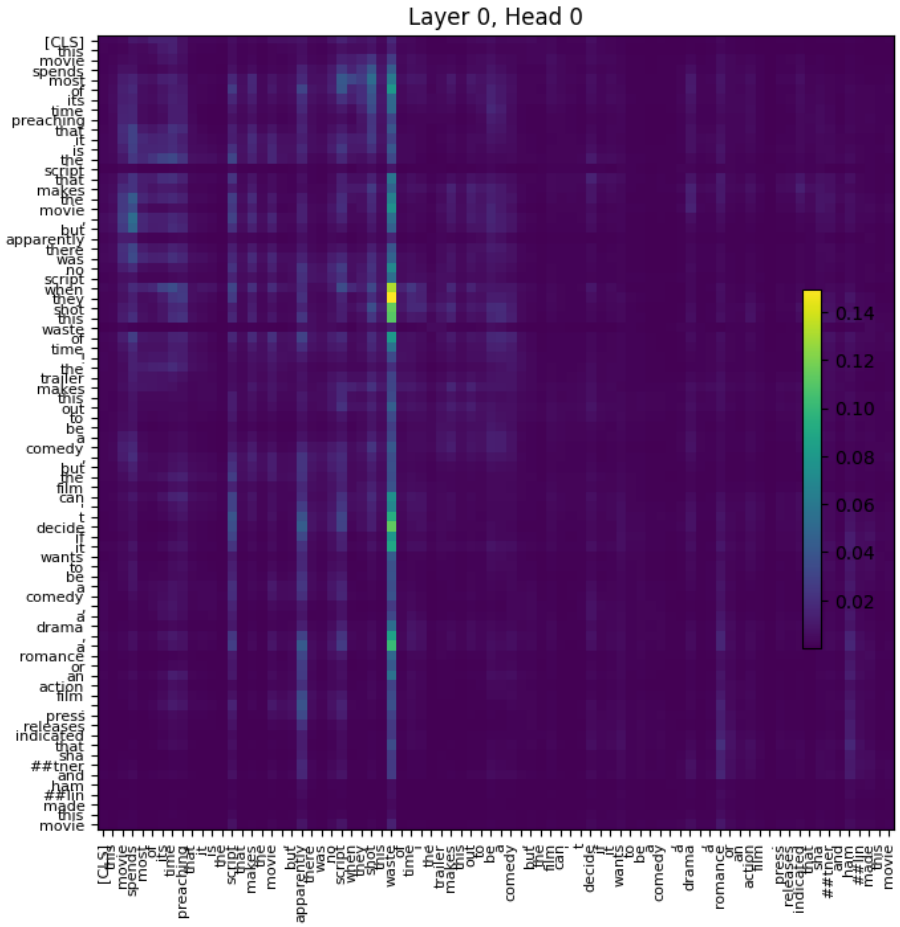


Figure 6 – Example1 - Attention head scores with associated tokens for single-head model

Pairwise cosine similarity between heads within multi-head variants is examined along with similarity between multi-head mean attention and single-head attention. The cosine similarity between the mean attention score for the multi-head and single-head variants for Example 1 is 0.89, indicating that the models are attending to largely the same patterns. The singular value analysis for Head 0 (Table 2) shows a prominent σ1 with quickly decaying subsequent values, suggesting a single dominant dimension, consistent with the primary single-word feature.

Head Number	**Singular Values, σ** (top 10)
1	*3.1801, 0.7259, 0.4521, 0.3400, 0.3350, 0.2664, 0.2016, 0.1842, 0.1597, 0.1153*
2	*1.9105, 0.8164, 0.5225, 0.3507, 0.3138, 0.2575, 0.2297, 0.1913, 0.1785, 0.1552*
3	*1.0471, 0.3567, 0.3180, 0.2600, 0.2079, 0.1537, 0.1274, 0.1035, 0.0889, 0.0792*
4	*1.9793, 0.6502, 0.3995, 0.2988, 0.2739, 0.2178, 0.2024, 0.1350, 0.1210, 0.1016*

Table 2 – Singular Values for Example 1 across the 4-head model

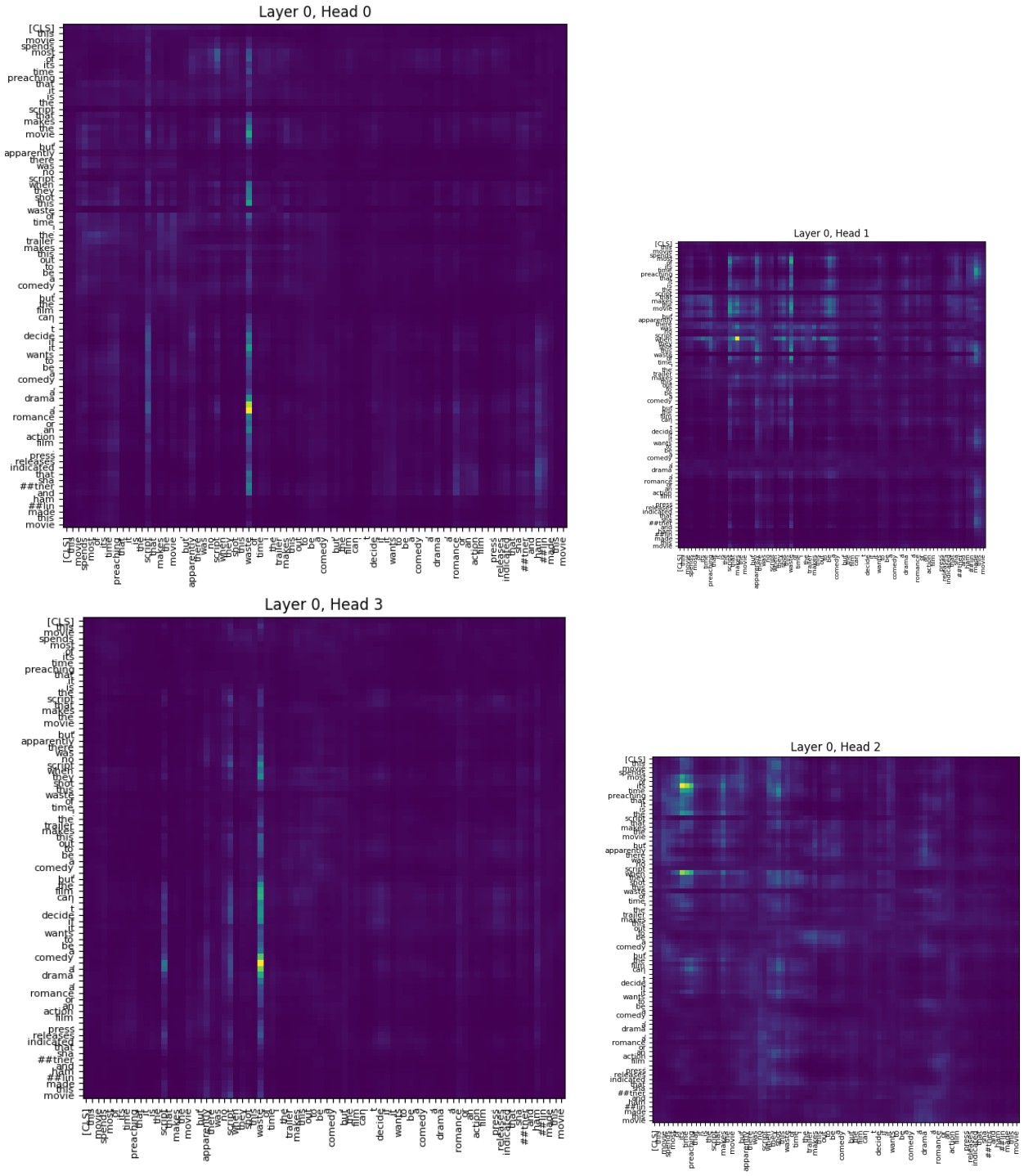


Figure 7,8,9,10 – Example 1 - Attention head scores with associated tokens for 4-head model

However, in some cases a single representational space may struggle to capture multiple conflicting dependencies simultaneously, motivating multi-head attention, as demonstrated in Example 2, where the single-head model produces a false negative (Figures 11-15). The attention matrix for the single-head model attends strongly to the token "*dreadful"* with other groups of negatively connoted tokens, such as "*who certainly hadn't watched or listened to it"*, *"sad thing is that"*. The multi-head model (4 heads) exhibits a similar pattern with head 1 anchoring on the phrase "*who certainly hadn't watched or listened to it"*, but its attention distribution spans longer vertical and horizontal dependencies, suggesting an incorporation of

broader contextual relationships that contribute to a true positive prediction. Other heads operate in different representational spaces - Head 2 performing a positional function, attending to the phrase *"this on thames television and loved it"*. The singular values for the multi-head model do not display a single dominant σ1, (Table 3), implying a more even distribution of attention across dimensions, supporting the visual analysis that the heads utilize multiple subspaces, capturing more diverse patterns that the single-head model cannot. The maximum cosine similarity between heads is 0.53, suggesting some correlation exists, but that heads largely operate in different subspaces (Figure 16).

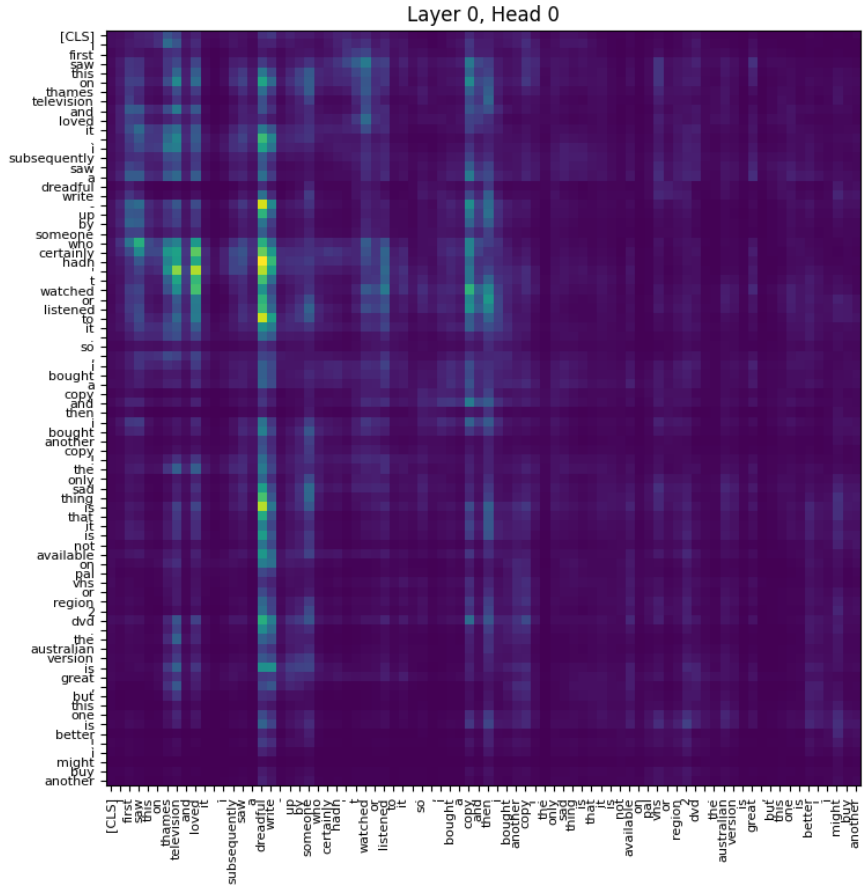
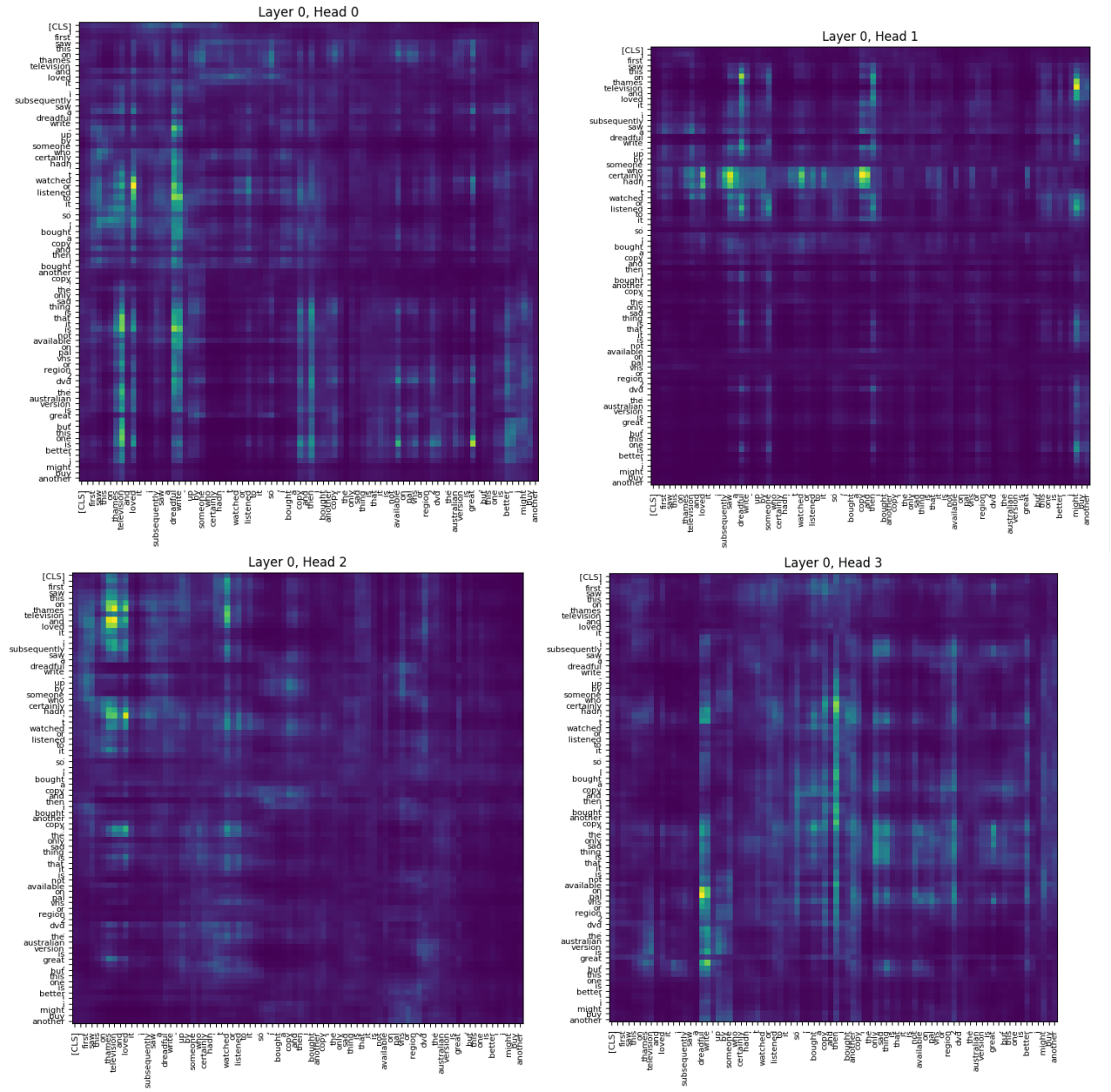


Figure 11 – Example 2, Attention head scores with associated tokens for single-head model

Head Number	**Singular Values, σ** (top 10)
1	*1.8638, 0.5696, 0.5436, 0.3836, 0.3119, 0.2155, 0.1632, 0.1463, 0.1278, 0.1171*
2	*1.6623, 0.7387, 0.4860, 0.4095, 0.2888, 0.2699, 0.2411, 0.1667, 0.1495, 0.1349*
3	*1.0505, 0.3592, 0.3165, 0.2598, 0.2126, 0.1567, 0.1271, 0.1062, 0.0877, 0.0778*
4	*1.3586, 0.3674, 0.3579, 0.2281,0.2008, 0.1897, 0.1455, 0.1206, 0.1121, 0.0911*

Table 3 – Singular Values for Example 2 across the 4-head model



Figures 12,13,14,15 – Example 2 - Attention head scores with associated tokens for 4-head model

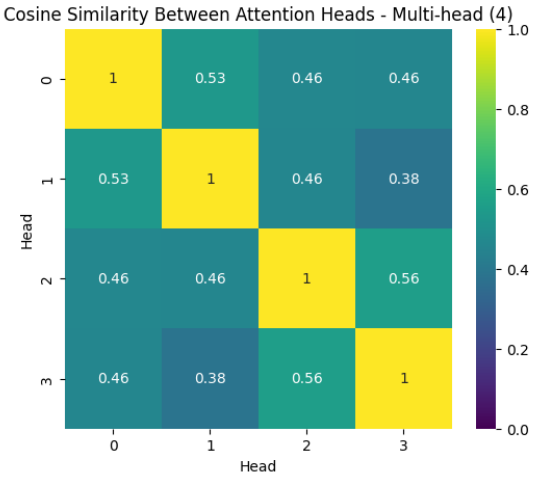


Figure 16 – Example 2 - Cosine similarity for the 4-head model

Analysis of rank and singular values provides insight into effective head dimensionality. Calculating effective rank depends heavily on the tolerance used to determine the contribution of a singular value to the matrix. Hence the values offer only an approximate indication of the matrix's effective dimensionality. However, a higher effective rank in the single-head model occurs, with the effective rank of the matrices reducing with more heads– producing sparser matrices. The 2-layer model maintains a higher effective rank across its total eight heads than 8 heads in a single layer (Table 4).

Example	Single-head model	4-head model	8-head model	4-head, 2 layer model
1	131	77,100,113,82	32, 52, 52, 48, 46, 51,49, 45	[114, 93, 103,103], [127, 94, 125, 104]
2	144	94,104,112,98	44, 50, 51, 51, 45, 49, 48, 45	[113,100,101,102], [134,101,128,105]

Table 4 – Effective rank of attention heads for two examples across models (tolerance = 0.001)

Conclusion

The experiments highlight how transformer performance and behaviour vary across attention head configurations. While multi-head attention can increase representational capacity, adding more heads does not always improve performance and may introduce redundancy depending on the task.

Although the models were not heavily optimised heavily for accuracy, they still performed well on the binary classification task. Smaller, non-deep-learning classifiers have been shown to perform competitively on the IMDB binary classification task (Narayanan et al., 2013), and the comparable performance of a single-head transformer to multi-head variants suggests that many attention heads may be underutilised or encode overlapping information. This interpretation was reinforced by strong overfitting observed when training on smaller subsets of the dataset without regularisation, particularly for larger models. The earlier convergence of single-head and 4-head variants (Figures 1,2) reinforces the idea that simpler patterns suffice for the task and may reflect the tendency of neural networks to learn lower-complexity patterns earlier in training, a phenomenon related to spectral bias in deep learning models (Rahaman et al., 2019).
In larger models, particularly the 8-head and 2-layer variants, reduced validation accuracy (Table 1) suggest that some attention heads may provide overlapping representations, a pattern further supported by inspection of attention matrices across variants. In Example 1, a single representational space suffices for the task, whereas in Example 2, multiple representational spaces improve performance. However, visual inspection across additional examples indicates that rarely more than 2-3 heads contribute largely independent information, often corroborated by the cosine similarity of the heads.

Several studies (Amsell et al.,2024, Cordonnier et al., 2020, Dong et al., 2023, Koveleva et al., 2019, Michel et al., 2019, Voita et al., 2019) use spectral analysis to understand the representational space in which attention heads operate. In this assignment examining cosine similarity, singular values and rank for selected examples provided some supporting evidence of the redundancy that may be being observed on this task. Rank decay tends to increase with model depth and pruning methods have been a popular subject of research (Dong et al., 2023, Michel et al., 2019, Voita et al., 2019). Since attention matrices change in response to input, a

more comprehensive spectral analysis for these models would examine similarity, rank and singular values across all samples, and would be the logical extension for further work. Focusing on more challenging tasks, such as language modelling where deeper models excel would likely reveal richer representational spaces and more complex relationships across varying attention head functions as detailed by Pande et al., (2021). Ablation methods could serve as an additional approach to more accurately quantify dimensionality in heads.

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